Full name: Ha Hoang Phuc Student: 222683 Class: **DH22TIN07 LESSON 1: Library management -** system - Identify candidate classes (Books, Readers, Loans, Employees, Publishers) + Books: main entity, book information management.

+ DocGia: person who borrows books.

+ PhieuMuon: records borrowing/returning.

+ NhanVien: system manager (librarian, warehouse keeper...).

+ NhaXuatBan: book source information.

- Properties and behavior (methods) of each class + Book o Attributes: maSach (PK, string), nameSach (string), tacGia (string), namXuatBan (int), maNXB (FK), soLuongTon (int), trangThai (string: "Available"/"On loan").

○ Behavior: addSach(), capNhatSach(), deleteSach(), saveSach(), capNhatTonKho().

+ DocGia ○ Attributes: maDocGia (PK, string), hoTen (string), daySinh (date), diaChi (string), soDienThoai (string), dayDangKy (date), gioiTinh (boolean).

○ Behavior: dangKyThe(), capNhatThongTin(), viewLichSuMuon(). + PhieuMuon o Attributes: maPhieu (PK, string), maDocGia (FK), maSach (FK), maNhanVien (FK), nhaMuon (date), ngoiHenTra (date), nhaTraThucTe (date), thanhTrang (string: "On loan"/"Paid"/"Overdue").

○ Behavior: createPhieuMuon(), searchSach(), saveTraQuaHan(), crystalPhieuPhat(). + NhanVien o Attributes: maNhanVien (PK, string), hoTen (string), chucVu (string), date of birth (date), soDienThoai (string).

○ Behavior: dangNhap (), quanLyDocGia (), quanLyMuonTra (), quanLySach ().().

+ NhaXuatBan ○ Attributes: nameNXB (PK, string), nameNXB (string), address (string), email (string).

○ Behavior: addNXB (), capNhatNXB ().

- Relationships: Books and Publishers: 1 book – 1 publisher

- Readers and Loans: 1 reader – many loans
- Books and Loans: 1 book – many loans
- Employees and Loans: 1 employee – many loans

LESSON 2: Online sales management system

- Identify main use cases

Main actors:

+ Customer (Customers / Potential Customers).
+ NguoiQuanLy / Admin (Administrator / Sales staff).

Main use cases (customer registration, ordering, payment, product management):

KhachHang:

- DangKyTaiKhan (Customer registration).
- DangNhap (Sign in).
- See SanPham (Search / View product list).
- ThemVaoGioHang (Add to cart).
- DatHang (Order).
- ThanhToan (Online payment / COD).
- View DonHang (View order history).

Admin / NguoiQuanLy:

- QuanLySanPham (Add/Edit/Delete products).
- QuanLyDonHang (View/Confirm/Cancel order).
- QuanLyKhachHang (Customer information management).
- QuanLyThanhToan (Payment processing, refund if necessary).

Extend / Include:

- ThanhToan includes XacNhanDonHang.
- DatHang extends ThanhToan (if paying online).

- From use case, find necessary classes

- Customer (Customer): from register, view orders.
- SanPham (Product): product management.
- GioHang (Cart): temporarily store products before ordering.
- Don Hang (Order): from order, payment.
- ChiTietDonHang (OrderDetail): product details in the order.
- ThanhToan (Payment): payment processing (may have subclasses such as OnlinePayment, COD).
- Admin or Personnel (inherited from User if there is a user system).

- Analyze the relationship between □classes (specify inheritance if □any)□ Customer and Customer: 1 customer – many □orders (abstract class) ←inherits □←KhachHang and Admin (KhachHang and □Admin are both the same type of User, sharing the same account, password, full name).

Lesson 3: Convert class diagrams into data tables

1. MonHoc table

No	School name	Data Types	Bind			
			PK	FK	Not NULL	Unique
1	maMon	Nvarchar(10)	x		x	x
2	tenMon	Nvarchar(50)			x	
3	soTinChiLT	Int			x	
4	soTinChiTH	Int			x	

2. Science Table

No	School name	Data Types	Bind			
			PK	FK	Not NULL	Unique
1	maKhoaHoc	Nvarchar(10)	x			x
2	name KhoaHoc	Nvarchar(50)			x	
3	ngayTao	Date			x	